DAILY PRACTICE CALENDAR

Grade 11 Math: Algebra 2 • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **California Common Core standards for Algebra 2**, the usual Grade 11 course. Students who want an extra challenge can add **Algebra 2 Competition Math**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Transformations of Functions
2	Aug 23–Aug 27	Composite & Inverse Functions
3	Aug 30–Sep 3	Complex Numbers
4	Sep 6–Sep 10	Quadratics with Complex Solutions
5	Sep 13–Sep 17	Polynomial Operations & Identities
6	Sep 20–Sep 24	Polynomial Division & the Remainder Theorem
7	Sep 27–Oct 1	Zeros & the Factor Theorem
8	Oct 4–Oct 8	Graphs of Polynomial Functions
9	Oct 11–Oct 15	Solving Polynomial Equations
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Rational Exponents & Radicals
12	Nov 1–Nov 5	Radical Equations
13	Nov 8–Nov 12	Rational Expressions
14	Nov 15–Nov 19	Rational Equations
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Rational Functions & Asymptotes
16	Dec 6–Dec 10	Exponential Functions & e
17	Dec 13–Dec 17	Logarithms
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Properties of Logarithms
19	Jan 10–Jan 14	Exponential & Logarithmic Equations
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Radians & the Unit Circle
22	Jan 31–Feb 4	Trig Values in All Four Quadrants
23	Feb 7–Feb 11	Graphs of Sine & Cosine
24	Feb 14–Feb 18	Modeling with Trigonometric Functions
25	Feb 21–Feb 25	Trig Identities & Equations
26	Feb 28–Mar 3	Sequences & Series
27	Mar 6–Mar 10	Geometric Series
28	Mar 13–Mar 17	The Normal Distribution
29	Mar 20–Mar 24	Surveys, Studies & Experiments
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Margin of Error & Inference
31	Apr 10–Apr 14	Choosing a Model
32	Apr 17–Apr 21	Year-End Review: Algebra & Functions
33	Apr 24–Apr 28	Year-End Review & Practice Test
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027 — Unit 1: Functions & Complex Numbers

Transformations of Functions

Core: $f(x-h)+k$ shifts a graph, $a \cdot f(x)$ stretches it, and $-f(x)$ flips it.

Mon Aug 16

Shifts

TRY Describe how $g(x) = (x+2)^2 - 5$ is moved from $y = x^2$.

Tue Aug 17

Reflections & stretches

TRY Describe how $g(x) = -3|x|$ changes $y = |x|$.

Wed Aug 18

Writing transformations

TRY Shift $y = \sqrt{x}$ right 4 and up 1. Write the new function.

Thu Aug 19

Even & odd functions

TRY Is $f(x) = x^3 - x$ even, odd or neither?

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027 — Unit 1: Functions & Complex Numbers

Composite & Inverse Functions

Core: An inverse function undoes the original: $f(f^{-1}(x)) = x$.

Mon Aug 23

Composition

TRY If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $f(g(2))$.

Tue Aug 24

Finding inverses

TRY Find the inverse of $f(x) = \frac{x-5}{2}$.

Wed Aug 25

Checking inverses

TRY Are $f(x) = 3x + 6$ and $g(x) = \frac{x}{3} - 2$ inverses?

Thu Aug 26

Restricting the domain

TRY Find the inverse of $f(x) = x^2$ for $x \geq 0$.

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027 — Unit 1: Functions & Complex Numbers

Complex Numbers

Core: $i^2 = -1$; complex numbers have the form $a + bi$.

Mon Aug 30

Square roots of negatives

TRY Simplify $\sqrt{-49}$.

Tue Aug 31

Multiplying

TRY Compute $(2+3i)(4-i)$.

Wed Sep 1

Dividing

TRY Compute $\dfrac{3+i}{1-i}$.

Thu Sep 2

Powers of i

TRY Find i^{50}.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027 — Unit 1: Functions & Complex Numbers

Quadratics with Complex Solutions

Core: When $b^2 - 4ac < 0$, the solutions are a complex-conjugate pair.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Square roots

TRY Solve $x^2 + 16 = 0$.

Tue Sep 7

Quadratic formula

TRY Solve $x^2 - 2x + 5 = 0$.

Wed Sep 8

The discriminant

TRY Describe the solutions of $3x^2 - 4x + 2 = 0$.

Thu Sep 9

Building from roots

TRY Write a quadratic $x^2 + bx + c$ with roots $3 \pm i$.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Polynomial Functions

Polynomial Operations & Identities

Core: Multiply carefully and look for patterns you can reuse.

Mon Sep 13
Multiplying
TRY Expand $(x-2)(x^2+2x+4)$.

Tue Sep 14
Cubing a binomial
TRY Expand $(x+1)^3$.

Wed Sep 15
Polynomial identities
TRY Use $(x^2-y^2)^2+(2x\cdot y)^2=(x^2+y^2)^2$ with $x=2$, $y=1$ to get a Pythagorean triple.

Thu Sep 16
Pascal's triangle
TRY Find the coefficient of x^2y^2 in $(x+y)^4$.

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Polynomial Functions

Polynomial Division & the Remainder Theorem

Core: The remainder when $P(x)$ is divided by $x-a$ is $P(a)$.

Mon Sep 20
Long division
TRY Divide $2x^3-3x^2-11x+6$ by $x-3$.

Tue Sep 21
Synthetic division
TRY Divide x^3-7x+6 by $x-1$.

Wed Sep 22
Remainder theorem
TRY Find the remainder when x^4-3x^2+2x-5 is divided by $x+2$.

Thu Sep 23
Finding k
TRY Find k so that $x-2$ is a factor of x^3-kx+4.

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Polynomial Functions

Zeros & the Factor Theorem

Core: $x-a$ is a factor of $P(x)$ exactly when $P(a)=0$.

Mon Sep 27
Factor theorem
TRY Is $x+1$ a factor of x^3+2x^2-x-2?

Tue Sep 28
Finding all zeros
TRY Find all zeros of x^3+2x^2-x-2.

Wed Sep 29
Rational Root Theorem
TRY List the possible rational zeros of $2x^3+x-6$.

Thu Sep 30
Conjugate zeros
TRY A cubic with real coefficients has zeros 2 and $1+i$. Find the third zero.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Polynomial Functions

Graphs of Polynomial Functions

Core: Degree and leading coefficient set the end behavior; multiplicity sets touch or cross.

Mon Oct 4
End behavior
TRY Describe the end behavior of $y=-2x^5+x$.

Tue Oct 5
Multiplicity
TRY Does $y=(x-1)^2(x+3)$ touch or cross the x-axis at $x=1$?

Wed Oct 6
Turning points
TRY What is the greatest number of turning points a degree-4 polynomial can have?

Thu Oct 7
Sketching clues
TRY Find the zeros and y-intercept of $y=x(x-2)(x+3)$.

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Polynomial Functions

Solving Polynomial Equations — **Core:** Factor by patterns, grouping or quadratic form.

Mon Oct 11
Sum of cubes
TRY Factor $x^3 + 27$.

Tue Oct 12
Quadratic form
TRY Solve $x^4 - 5x^2 + 4 = 0$.

Wed Oct 13
Grouping
TRY Solve $x^3 - 2x^2 - 9x + 18 = 0$.

Thu Oct 14
Complex solutions
TRY Find all solutions of $x^3 - 8 = 0$.

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Polynomial Functions

Review & Checkpoint 1 — **Core:** Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: inverses
TRY If $f(x) = 4x - 1$, find $f^{-1}(7)$.

Tue Oct 19
Review: complex numbers
TRY Compute $(1 + 2i)^2$.

Wed Oct 20
Review: remainders
TRY Find the remainder when $x^3 - 2x + 1$ is divided by $x - 2$.

Thu Oct 21
Review: zeros
TRY Solve $x^3 - 4x = 0$.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Radical & Rational Functions

Rational Exponents & Radicals — **Core:** $a^{m/n} = (\sqrt[n]{a})^m$.

Mon Oct 25
Evaluating
TRY Compute $16^{3/4}$.

Tue Oct 26
Simplifying radicals
TRY Simplify $\sqrt[3]{54x^4}$.

Wed Oct 27
Exponent rules
TRY Simplify $x^{1/2} \cdot x^{1/3}$.

Thu Oct 28
Radical functions
TRY Find the domain of $f(x) = \sqrt{2x - 6} + 1$.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Radical & Rational Functions

Radical Equations — **Core:** Isolate the radical, raise both sides to a power, and check for extraneous solutions.

Mon Nov 1
Square-root equations
TRY Solve $\sqrt{x + 5} = 4$.

Tue Nov 2
Extraneous solutions
TRY Solve $\sqrt{x + 6} = x$.

Wed Nov 3
Cube-root equations
TRY Solve $\sqrt[3]{2x - 1} = 3$.

Thu Nov 4
Rational exponents
TRY Solve $x^{2/3} = 9$.

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Radical & Rational Functions

Rational Expressions **Core:** Factor first, then cancel common factors; never cancel terms.

Mon Nov 8
Simplifying
TRY Simplify $\dfrac{x^2-9}{x^2+5x+6}$.

Tue Nov 9
Dividing
TRY Simplify $\dfrac{x}{x+1} \div \dfrac{x^2}{x^2-1}$.

Wed Nov 10
Adding
TRY Simplify $\dfrac{1}{x}+\dfrac{2}{x+1}$.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Complex fractions
TRY Simplify $\dfrac{1+\frac{1}{x}}{1-\frac{1}{x}}$.

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Radical & Rational Functions

Rational Equations **Core:** Multiply by the LCD, then check for excluded values.

Mon Nov 15
Proportions
TRY Solve $\dfrac{3}{x-2}=\dfrac{5}{x+2}$.

Tue Nov 16
Using the LCD
TRY Solve $\dfrac{1}{x}+\dfrac{1}{2x}=\dfrac{3}{4}$.

Wed Nov 17
Extraneous solutions
TRY Solve $\dfrac{x}{x-3}=\dfrac{3}{x-3}+2$.

Thu Nov 18
Work problems
TRY Pipe A fills a tank in 6 hours and pipe B in 3 hours. How long do they take together?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Radical & Rational Functions

Rational Functions & Asymptotes **Core:** Vertical asymptotes come from zeros of the denominator that do not cancel.

Mon Nov 29
Vertical asymptotes
TRY Find the vertical asymptote of $f(x)=\dfrac{x+1}{x-4}$.

Tue Nov 30
Horizontal asymptotes
TRY Find the horizontal asymptote of $f(x)=\dfrac{3x^2+1}{x^2-4}$.

Wed Dec 1
Holes
TRY Where is the hole in the graph of $f(x)=\dfrac{x^2-1}{x-1}$?

Thu Dec 2
Inverse variation
TRY y varies inversely with x, and $y=6$ when $x=4$. Find y when $x=8$.

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Exponential & Logarithmic Functions

Exponential Functions & e **Core:** $f(x)=a\cdot b^x$; continuous growth uses $e\approx 2.718$.

Mon Dec 6
Growth rate
TRY By what percent does $y=3(1.07)^x$ grow each step?

Tue Dec 7
Compound interest
TRY $1,000 earns 6% a year, compounded monthly. How much is there after 1 year?

Wed Dec 8
Continuous growth
TRY $500 earns 4% compounded continuously. How much is there after 10 years?

Thu Dec 9
Asymptotes
TRY Find the horizontal asymptote of $y=2^x-3$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Exponential & Logarithmic Functions

Logarithms **Core:** $\log_b x = y$ means $b^y = x$: a logarithm is an exponent.

Mon Dec 13
Converting forms
TRY Write $2^5 = 32$ in logarithmic form.

Tue Dec 14
Evaluating
TRY Find $\log_3 81$ and $\log 0.01$.

Wed Dec 15
Natural logs
TRY Find $\ln e^3$ and $e^{\ln 7}$.

Thu Dec 16
Log graphs
TRY Find the domain of $y = \log(x - 2)$.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Exponential & Logarithmic Functions

Properties of Logarithms **Core:** Logs turn products into sums, quotients into differences and powers into multiples.

Mon Jan 3
Quotient rule
TRY Compute $\log_2 24 - \log_2 3$.

Tue Jan 4
Expanding
TRY Expand $\log(x^3 y^2)$.

Wed Jan 5
Condensing
TRY Write $2\ln x - \ln y$ as a single logarithm.

Thu Jan 6
Change of base
TRY Use change of base to estimate $\log_5 20$.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Exponential & Logarithmic Functions

Exponential & Logarithmic Equations **Core:** Match bases when you can; otherwise take a log of both sides.

Mon Jan 10
Same base
TRY Solve $4^{x+1} = 32$.

Tue Jan 11
Using logs
TRY Solve $3^x = 20$.

Wed Jan 12
Log equations
TRY Solve $\log_2(x + 3) = 4$.

Thu Jan 13
Doubling time
TRY Money earns 5% a year, compounded yearly. About how long until it doubles?

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Exponential & Logarithmic Functions

Mid-Year Review & Checkpoint **Core:** Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: rational equations
TRY Solve $\frac{2}{x} = \frac{3}{x - 1}$.

Tue Jan 18
Review: rational exponents
TRY Compute $27^{2/3}$.

Wed Jan 19
Review: logarithms
TRY Find $\log_4 64$.

Thu Jan 20
Review: exponential models
TRY 200 g of a substance has a half-life of 10 years. How much is left after 30 years?

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Trigonometric Functions

Radians & the Unit Circle

Core: π radians $= 180°$; on the unit circle, $(\cos\theta, \sin\theta)$.

Mon Jan 24
Converting
TRY Convert $150°$ to radians and $\frac{3\pi}{2}$ to degrees.

Tue Jan 25
Arc length
TRY Find the arc length for a central angle of $\frac{\pi}{3}$ in a circle of radius 4.

Wed Jan 26
Coterminal angles
TRY Find the angle between 0 and 2π that is coterminal with $-\frac{\pi}{4}$.

Thu Jan 27
Reference angles
TRY Find the reference angle of $210°$.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Trigonometric Functions

Trig Values in All Four Quadrants

Core: Use reference angles, then attach the sign for the quadrant.

Mon Jan 31
Unit circle values
TRY Find $\sin\frac{5\pi}{6}$ and $\cos\frac{5\pi}{6}$.

Tue Feb 1
Signs by quadrant
TRY If $\sin\theta < 0$ and $\cos\theta > 0$, which quadrant is θ in?

Wed Feb 2
From one ratio
TRY If $\sin\theta = -\frac{3}{5}$ and θ is in Quadrant III, find $\cos\theta$ and $\tan\theta$.

Thu Feb 3
Pythagorean identity
TRY If $\cos\theta = \frac{5}{13}$ and θ is in Quadrant IV, find $\sin\theta$.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Trigonometric Functions

Graphs of Sine & Cosine

Core: For $y = a\sin(b(x-h)) + k$: amplitude $|a|$, period $\frac{2\pi}{b}$, midline $y = k$.

Mon Feb 7
Amplitude
TRY Find the amplitude of $y = 3\sin x$.

Tue Feb 8
Period
TRY Find the period of $y = \cos 2x$.

Wed Feb 9
Midline
TRY Find the midline, maximum and minimum of $y = 2\sin x + 5$.

Thu Feb 10
Phase shift
TRY How is $y = \sin\left(x - \frac{\pi}{4}\right)$ moved from $y = \sin x$?

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Trigonometric Functions

Modeling with Trigonometric Functions

Core: Repeating patterns — tides, wheels, seasons — can be modeled with sine and cosine.

Mon Feb 14
Writing a function
TRY Write a cosine function with amplitude 4, period 2π and midline $y = 1$.

Tue Feb 15
Period from b
TRY Find the period of $y = \sin\left(\frac{\pi x}{6}\right)$.

Wed Feb 16
Ferris wheels
TRY A Ferris wheel has diameter 40 m, and its center is 25 m above the ground. Find the greatest and least heights of a seat.

Thu Feb 17
Tides
TRY High tide is 9 ft and low tide is 1 ft. Find the amplitude and the midline.

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Trigonometric Functions

Trig Identities & Equations

Core: $\sin^2\theta + \cos^2\theta = 1$; solve trig equations on the unit circle.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Pythagorean identity

TRY Simplify $1 - \cos^2\theta$.

Tue Feb 22

Quotient identity

TRY Simplify $\tan\theta \cdot \cos\theta$.

Wed Feb 23

Solving

TRY Solve $2\cos x = 1$ for $0 \le x < 2\pi$.

Thu Feb 24

Squared equations

TRY Solve $\sin^2 x = \frac{1}{4}$ for $0 \le x < 2\pi$.

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 6: Sequences, Series & Statistics

Sequences & Series

Core: An arithmetic series adds to (number of terms) × (first + last) ÷ 2.

Mon Feb 28

Arithmetic sequences

TRY If $a_1 = 3$ and $d = 4$, find a_{20}.

Tue Feb 29

Arithmetic series

TRY Find the sum of the first 20 terms of that sequence.

Wed Mar 1

Sigma notation

TRY Evaluate $\sum_{k=1}^{5}(2k+1)$.

Thu Mar 2

Recursive sequences

TRY If $a_1 = 2$ and $a_n = 3a_{n-1}$, find a_5.

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Sequences, Series & Statistics

Geometric Series

Core: Finite: $S_n = a_1\frac{1-r^n}{1-r}$. Infinite, with $|r| < 1$: $S = \frac{a_1}{1-r}$.

Mon Mar 6

nth term

TRY Find a_8 for 5, 10, 20, . . .

Tue Mar 7

Finite series

TRY Find the sum of the first 8 terms of 5, 10, 20, . . .

Wed Mar 8

Infinite series

TRY Find $12 + 6 + 3 + \cdots$.

Thu Mar 9

Bouncing balls

TRY A ball dropped from 10 ft bounces back to half its height each time. How far does it travel in all?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Sequences, Series & Statistics

The Normal Distribution

Core: About 68%, 95% and 99.7% of the data lie within 1, 2 and 3 standard deviations of the mean.

Mon Mar 13

The 68-95-99.7 rule

TRY Scores have mean 100 and SD 15. What percent are between 85 and 115?

Tue Mar 14

z-scores

TRY Find the z-score of 130 when the mean is 100 and the SD is 15.

Wed Mar 15

Percent above

TRY For the same scores, about what percent are above 130?

Thu Mar 16

Comparing scores

TRY Test A: 85 (mean 75, SD 5). Test B: 90 (mean 80, SD 10). Which score is better relative to its test?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Sequences, Series & Statistics

Surveys, Studies & Experiments

Core: Only a randomized experiment can show cause and effect.

Mon Mar 20
Types of studies
TRY Researchers randomly assign patients to a new drug or a placebo. Is this a survey, an observational study or an experiment?

Tue Mar 21
Bias
TRY A website asks its visitors to vote in a poll. What is the problem?

Wed Mar 22
Random samples
TRY How can you fairly choose 50 students from 1,000?

Thu Mar 23
Cause and effect
TRY Which kind of study can show that one thing causes another?

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Sequences, Series & Statistics

Margin of Error & Inference

Core: A sample statistic plus or minus the margin of error gives a likely range.

Mon Apr 3
Margin of error
TRY A poll reports 52% $\pm$ 3%. Give the interval.

Tue Apr 4
Sample size
TRY Estimate the margin of error $\frac{1}{\sqrt{n}}$ for a sample of $n = 400$.

Wed Apr 5
Interpreting
TRY From 52% $\pm$ 3%, can we be confident the candidate has a majority?

Thu Apr 6
Comparing treatments
TRY Group means differ by 3. In simulations, a difference of 3 or more happens by chance 2% of the time. Is the difference significant?

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Sequences, Series & Statistics

Choosing a Model

Core: Constant differences: linear. Constant second differences: quadratic. Constant ratios: exponential.

Mon Apr 10
Which model?
TRY Which kind of model fits y: 2, 6, 18, 54?

Tue Apr 11
Comparing growth
TRY Which grows fastest in the long run: x^3, 3^x or $300x$?

Wed Apr 12
Fitting an exponential
TRY Find $y = a \cdot b^x$ through $(0, 5)$ and $(2, 45)$.

Thu Apr 13
Interpreting a model
TRY By what percent does $P(t) = 2000(0.95)^t$ change each year?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Year-End Review

Year-End Review: Algebra & Functions

Core: Mix the skills from Units 1–4 and show your work.

Mon Apr 17
Review: polynomials
TRY Find all zeros of $x^3 - x^2 - 4x + 4$.

Tue Apr 18
Review: rational functions
TRY Find the vertical asymptotes of $f(x) = \dfrac{1}{x^2 - 9}$.

Wed Apr 19
Review: log equations
TRY Solve $\log x + \log(x - 3) = 1$.

Thu Apr 20
Timed set: 10 in 20 min
TRY Compute $(2 - i)(2 + i)$.

Fri Apr 21
Weekly Quiz
10 mixed review problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Year-End Review

Year-End Review & Practice Test

Core: Practice a full test, then learn from every error.

Mon Apr 24
Review: trigonometry
TRY Find $\cos \frac{4\pi}{3}$.

Tue Apr 25
Review: series
TRY Find $1 + \frac{1}{3} + \frac{1}{9} + \cdots$.

Wed Apr 26
Review: statistics
TRY Find the z-score of 70 when the mean is 60 and the SD is 4.

Thu Apr 27
Practice Test
Full-length Algebra 2 practice test covering the whole year.

Fri Apr 28
Error-log review
Redo every missed problem from the test; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Year-End Review

Final Checkpoint & Celebration

Core: Show what you have learned and get ready for Precalculus.

Mon May 1
Light review
TRY Compute $\log_2 8 + \log_3 9$.

Tue May 2
Mixed warm-up set
TRY Find the period of $y = 3 \sin 4x$.

Wed May 3
Final Checkpoint
Covers the whole year.

Thu May 4
Checkpoint review
Go over solutions together; students explain one problem each.

Fri May 5
Celebration & reflection
Certificates, progress review, and goals for Precalculus next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: Left 2, down 5
Tue: Reflected over the x-axis and stretched by 3
Wed: $y = \sqrt{x-4}+1$
Thu: Odd

Week 2
Mon: 2
Tue: $f^{-1}(x) = 2x+5$
Wed: Yes: $f(g(x)) = x$
Thu: $f^{-1}(x) = \sqrt{x}$

Week 3
Mon: $7i$
Tue: $11+10i$
Wed: $1+2i$
Thu: -1

Week 4
Mon: $x = \pm 4i$
Tue: $x = 1 \pm 2i$
Wed: Two complex (non-real) solutions
Thu: $x^2 - 6x + 10$

Week 5
Mon: $x^3 - 8$
Tue: $x^3 + 3x^2 + 3x + 1$
Wed: 3, 4, 5
Thu: 6

Week 6
Mon: $2x^2 + 3x - 2$
Tue: $x^2 + x - 6$
Wed: -5
Thu: $k = 6$

Week 7
Mon: Yes
Tue: $-2, -1, 1$
Wed: $\pm 1, \pm 2, \pm 3, \pm 6, \pm\frac{1}{2}, \pm\frac{3}{2}$
Thu: $1 - i$

Week 8
Mon: Up on the left, down on the right
Tue: Touches
Wed: 3
Thu: Zeros 0, 2, -3; y-intercept 0

Week 9
Mon: $(x+3)(x^2-3x+9)$
Tue: $x = \pm 1, \pm 2$
Wed: $x = 2, \pm 3$
Thu: 2 and $-1 \pm i\sqrt{3}$

Week 10
Mon: 2
Tue: $-3+4i$
Wed: 5
Thu: $x = 0, \pm 2$

Week 11
Mon: 8
Tue: $3x\sqrt[3]{2x}$
Wed: $x^{5/6}$
Thu: $x \geq 3$

Week 12
Mon: $x = 11$
Tue: $x = 3$ ($x = -2$ is extraneous)
Wed: $x = 14$
Thu: $x = \pm 27$

Week 13
Mon: $\frac{x-3}{x+2}$
Tue: $\frac{x-1}{x}$
Wed: $\frac{3x+1}{x(x+1)}$
Thu: $\frac{x+1}{x-1}$

Week 14
Mon: $x = 8$
Tue: $x = 2$
Wed: No solution ($x = 3$ is excluded)
Thu: 2 hours

Week 15
Mon: $x = 4$
Tue: $y = 3$
Wed: At $(1, 2)$
Thu: 3

Week 16
Mon: 7%
Tue: \$1,061.68
Wed: About \$745.91
Thu: $y = -3$

Week 17
Mon: $\log_2 32 = 5$
Tue: 4 and -2
Wed: 3 and 7
Thu: $x > 2$

Week 18
Mon: 3
Tue: $3\log x + 2\log y$
Wed: $\ln \frac{x^2}{y}$
Thu: About 1.861

Week 19
Mon: $x = \frac{3}{2}$
Tue: $x = \frac{\ln 20}{\ln 3} \approx 2.727$
Wed: $x = 13$
Thu: About 14.2 years

Week 20
Mon: $x = -2$
Tue: 9
Wed: 3
Thu: 25 g

Week 21
Mon: $\frac{5\pi}{6}$ and $270°$
Tue: $\frac{4\pi}{3}$
Wed: $\frac{7\pi}{4}$
Thu: $30°$

Week 22
Mon: $\frac{1}{2}$ and $-\frac{\sqrt{3}}{2}$
Tue: Quadrant IV
Wed: $-\frac{4}{5}$ and $\frac{3}{4}$
Thu: $-\frac{12}{13}$

Week 23
Mon: 3
Tue: π
Wed: $y = 5$; max 7, min 3
Thu: Right $\frac{\pi}{4}$

Week 24
Mon: $y = 4\cos x + 1$
Tue: 12
Wed: 45 m and 5 m
Thu: Amplitude 4; midline 5 ft

Week 25
Mon: $\sin^2\theta$
Tue: $\sin\theta$
Wed: $\frac{\pi}{3}, \frac{5\pi}{3}$
Thu: $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

Week 26
Mon: 79
Tue: 820
Wed: 35
Thu: 162

Week 27
Mon: 640
Tue: 1,275
Wed: 24
Thu: 30 ft

Week 28
Mon: About 68%
Tue: 2
Wed: About 2.5%
Thu: Test A ($z = 2$ vs. $z = 1$)

Week 29
Mon: An experiment
Tue: Voluntary response bias
Wed: A simple random sample (e.g., random numbers)
Thu: A randomized experiment

Week 30
Mon: 49% to 55%
Tue: 5%
Wed: No: the interval goes below 50%
Thu: Yes

Week 31
Mon: Exponential
Tue: 3^x
Wed: $y = 5 \cdot 3^x$
Thu: It decreases 5%

Week 32
Mon: 1, ± 2
Tue: $x = \pm 3$
Wed: $x = 5$
Thu: 5

Week 33
Mon: $-\frac{1}{2}$
Tue: $\frac{3}{2}$
Wed: 2.5

Week 34
Mon: 5
Tue: $\frac{\pi}{2}$

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